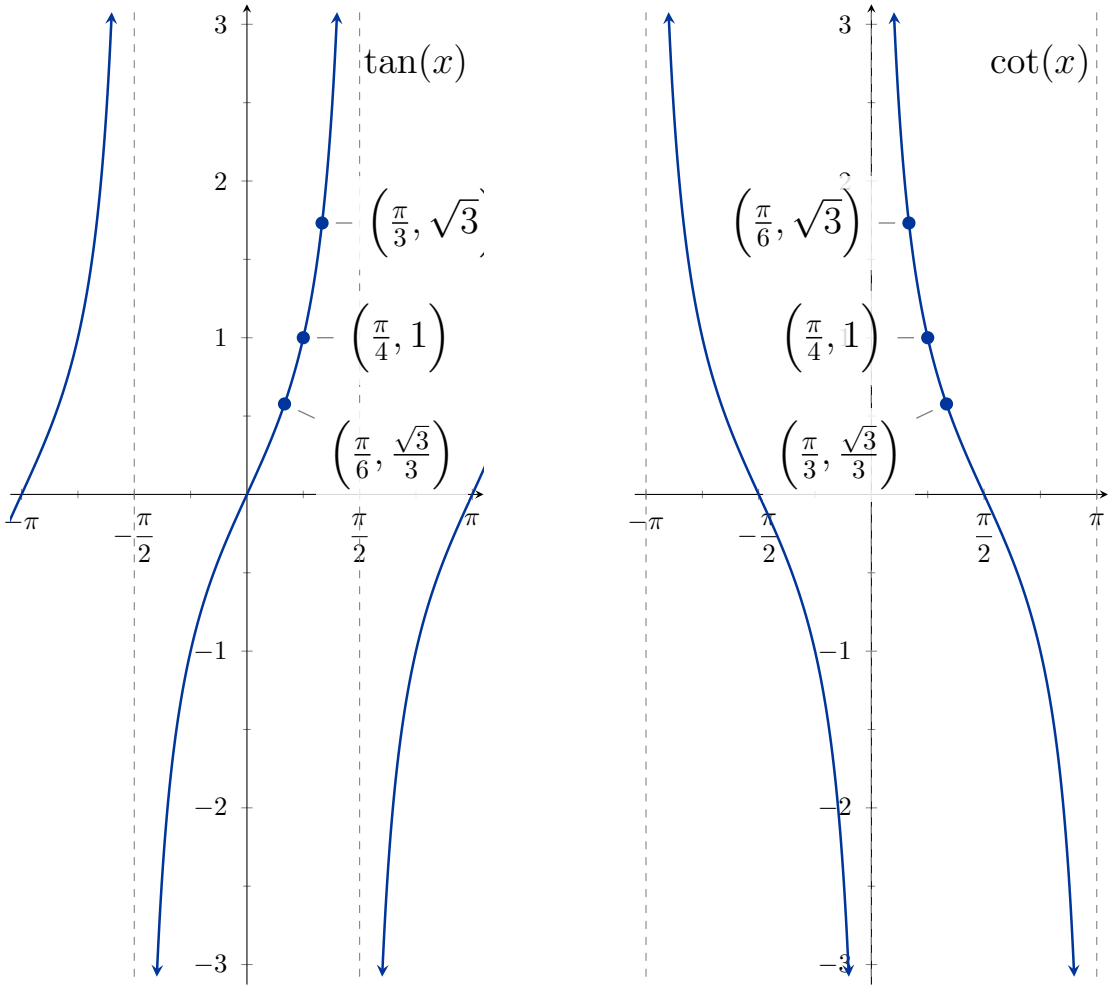
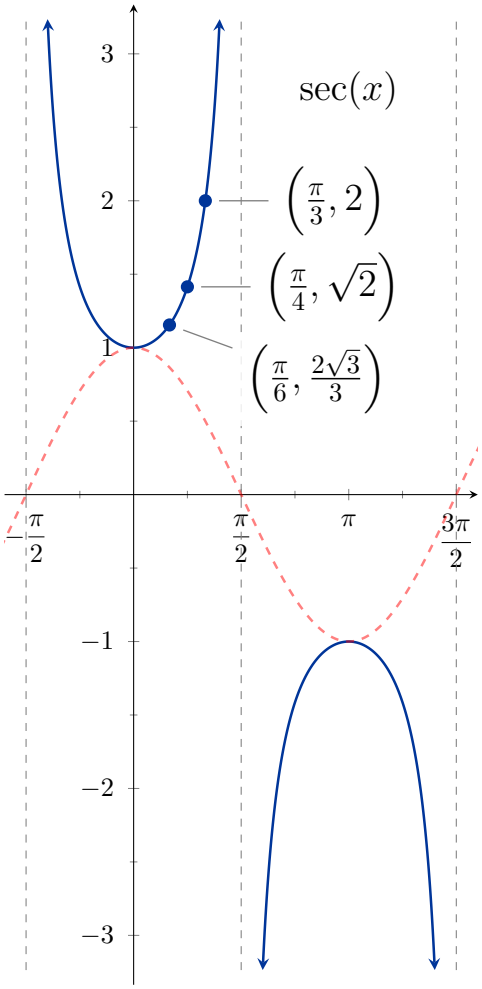
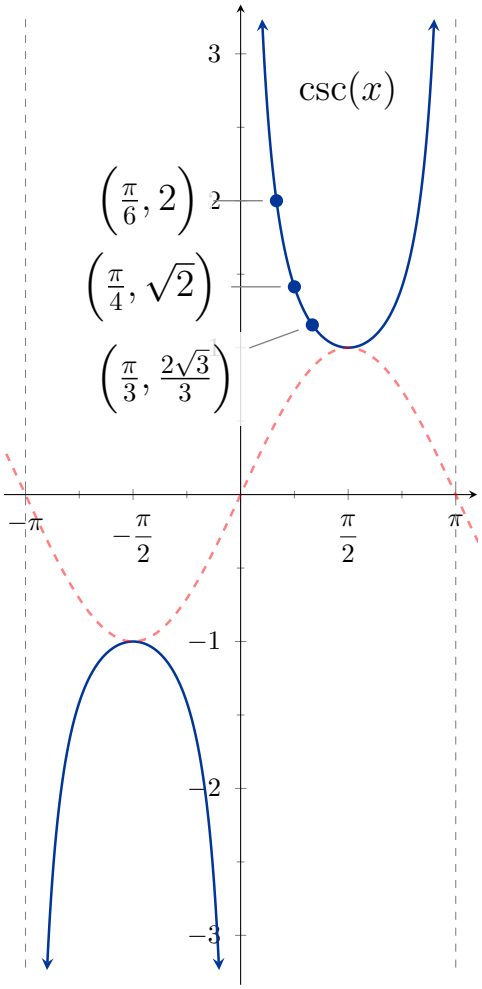
6.2: Graphs of the Other Trigonometric Functions




sec(x)
(π/3, 2)
(π/4, √2)
(π/6, 2√3/3)
3
2
1
−1
−2
−3
−π/2
π/2
π
3π/2


csc(x)
(π/6, 2)
(π/4, √2)
(π/3, 2√3/3)
3
2
−1
−2
−3
−π
−π/2
π/2
π

The general forms of the other trigonometric functions are

$$y = A\tan(Bx - C) + D, \qquad y = A\cot(Bx - C) + D,$$
$$y = A\sec(Bx - C) + D, \qquad y = A\csc(Bx - C) + D.$$

where

- A determines the scaling factor,
- B determines the period,
- C determines the phase shift, and
- D determines the vertical shift.

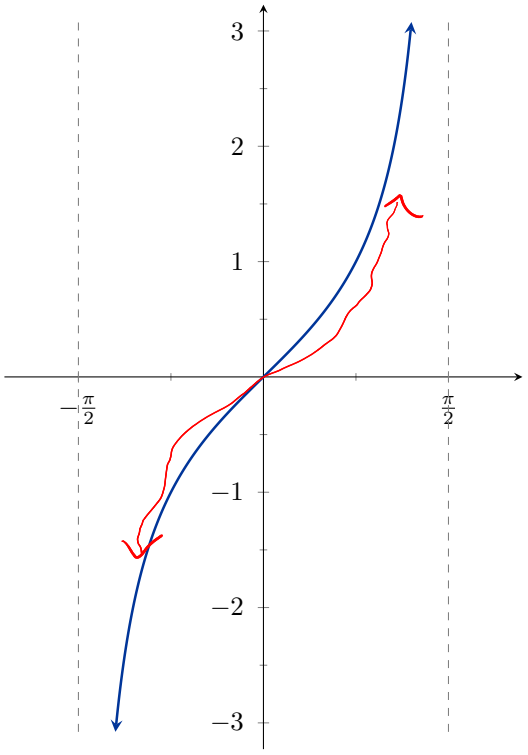
Example. Draw one period of the function $y = \frac{1}{2}\tan\left(\frac{\pi}{2}x\right)$

$A = 1/2$ ← scaling factor

$B = \pi/2$ ← Period $= \frac{\pi}{B} = 2$

$C = 0$

$D = 0$



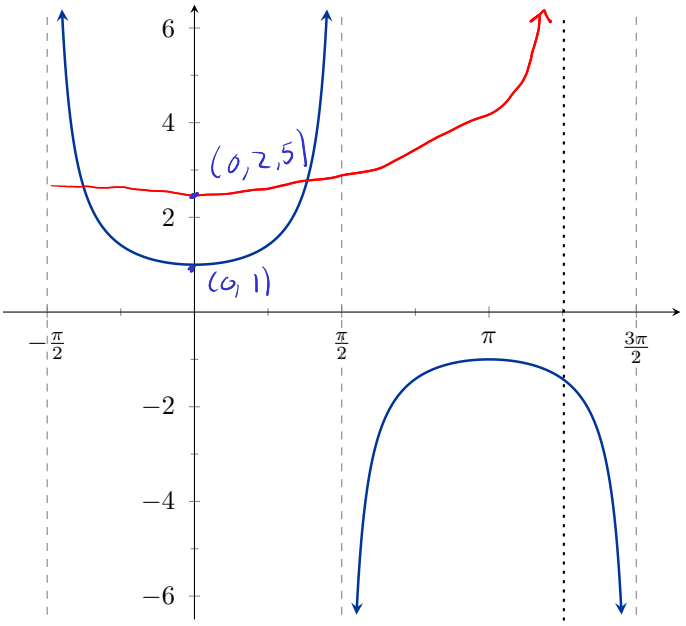
Example. Identify the domain, range, and draw one period of $y = 2.5 \sec\left(\frac{2}{5}x\right)$.

$A = 2.5 \leftarrow$ scaling factor

$B = \frac{2}{5} \leftarrow$ Period $= \frac{\pi}{B} = \frac{5\pi}{2}$

$C = 0$

$D = 0$



Domain $x \neq \frac{5\pi}{4} + n\frac{5\pi}{2}$, $n = 0, \pm 1, \pm 2, \ldots$

Range $(-\infty, -2.5] \cup [2.5, \infty)$

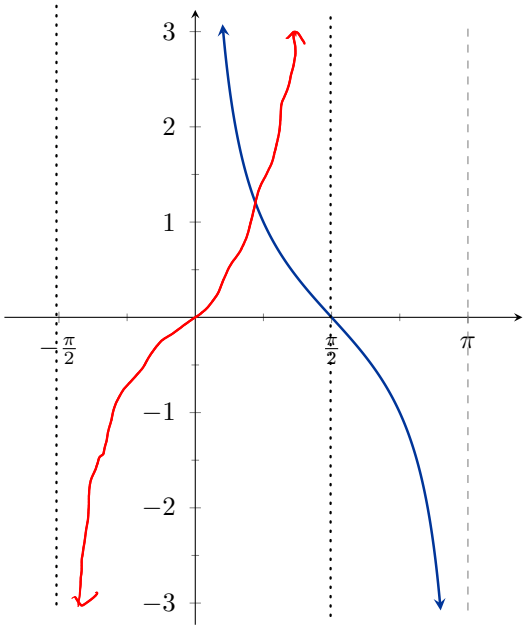
Example. Identify the domain, range, and draw one period of $y = -\cot\left(x + \frac{\pi}{2}\right)$.

$A = -1 \leftarrow$ scaling factor

$B = 1 \leftarrow$ Period $= \frac{\pi}{B} = \pi$

$C = -\frac{\pi}{2} \leftarrow$ Phase shift $= \frac{C}{B} = -\frac{\pi}{2}$

$D = 0$



Domain $x \neq \frac{\pi}{2} + n\pi$, $n = 0, \pm 1, \pm 2, \ldots$

Range $(-\infty, \infty)$